

Supplementary material on regression (QS 2025)

Useful results for matrix inversion

The **Woodbury formula** (or **Matrix inversion lemma**)

$$(\mathbf{A} + \mathbf{UBV})^{-1} = \mathbf{A}^{-1} - \mathbf{A}^{-1}\mathbf{UB}(\mathbf{B} + \mathbf{BVA}^{-1}\mathbf{UB})^{-1}\mathbf{BVA}^{-1}$$

Where the matrix dimensions are: $\mathbf{A}:(p \cdot p)$ $\mathbf{U}:(pq)$ $\mathbf{B}:(q \cdot q)$ $\mathbf{V}:(q \cdot p)$

and the assumptions are: $(\mathbf{B} + \mathbf{BVA}^{-1}\mathbf{UB})$ and $\mathbf{A}$ are invertible.

Verification

First note that: $(\mathbf{A} + \mathbf{UBV})\mathbf{A}^{-1}\mathbf{UB} = \mathbf{UB} + \mathbf{UBVA}^{-1}\mathbf{UB} = \mathbf{U}(\mathbf{B} + \mathbf{BVA}^{-1}\mathbf{UB})$

Then, we have to verify that: $(\mathbf{A} + \mathbf{UBV})\left(\mathbf{A}^{-1} + \mathbf{A}^{-1}\mathbf{UB}(\mathbf{B} + \mathbf{BVA}^{-1}\mathbf{UB})^{-1}\mathbf{BVA}^{-1}\right) = \mathbf{I}_p$

Let us start the verification.

$$\begin{aligned} & (\mathbf{A} + \mathbf{UBV})\left(\mathbf{A}^{-1} + \mathbf{A}^{-1}\mathbf{UB}(\mathbf{B} + \mathbf{BVA}^{-1}\mathbf{UB})^{-1}\mathbf{BVA}^{-1}\right) = \\ &= \mathbf{I}_p + \mathbf{UBVA}^{-1} - \mathbf{UB}(\mathbf{B} + \mathbf{BVA}^{-1}\mathbf{UB})^{-1}\mathbf{BVA}^{-1} - \mathbf{UBVA}^{-1}\mathbf{UB}(\mathbf{B} + \mathbf{BVA}^{-1}\mathbf{UB})^{-1}\mathbf{BVA}^{-1} = \\ &= \mathbf{I}_p + \mathbf{UBVA}^{-1} - \mathbf{U}\left[\mathbf{B}(\mathbf{B} + \mathbf{BVA}^{-1}\mathbf{UB})^{-1} + \mathbf{BVA}^{-1}\mathbf{UB}(\mathbf{B} + \mathbf{BVA}^{-1}\mathbf{UB})^{-1}\right]\mathbf{BVA}^{-1} = \\ &= \mathbf{I}_p + \mathbf{UBVA}^{-1} - \mathbf{U}\left[(\mathbf{B} + \mathbf{BVA}^{-1}\mathbf{UB})(\mathbf{B} + \mathbf{BVA}^{-1}\mathbf{UB})^{-1}\right]\mathbf{BVA}^{-1} = \mathbf{I}_p + \mathbf{UBVA}^{-1} - \mathbf{UBVA}^{-1} = \mathbf{I}_p \\ & \text{QED.} \end{aligned}$$

There are some useful special cases

Case1

Assume $p=q$ and $\mathbf{U} = \mathbf{V} = \mathbf{I}_p$

$$(\mathbf{A} + \mathbf{B})^{-1} = \mathbf{A}^{-1} - \mathbf{A}^{-1}(\mathbf{I}_p + \mathbf{BA}^{-1})\mathbf{BA}^{-1}$$

Verification

$$\begin{aligned} & (\mathbf{A} + \mathbf{B})^{-1} = \mathbf{A}^{-1} - \mathbf{A}^{-1}\mathbf{B}(\mathbf{B} + \mathbf{BA}^{-1}\mathbf{B})^{-1}\mathbf{BA}^{-1} = \mathbf{A}^{-1} - \mathbf{A}^{-1}\mathbf{B}\left[(\mathbf{I}_p + \mathbf{BA}^{-1})\mathbf{B}\right]^{-1}\mathbf{BA}^{-1} = \\ &= \mathbf{A}^{-1} - \mathbf{A}^{-1}\mathbf{BB}^{-1}(\mathbf{I}_p + \mathbf{BA}^{-1})^{-1}\mathbf{BA}^{-1} = \mathbf{A}^{-1} - \mathbf{A}^{-1}(\mathbf{I}_p + \mathbf{BA}^{-1})^{-1}\mathbf{BA}^{-1} \end{aligned}$$

since: $(\mathbf{AB})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1}$

Case 2

Assume $q=1$ $\mathbf{B} = \mathbf{I}_q = 1 \rightarrow \mathbf{U} = \mathbf{u}$ and $\mathbf{V} = \mathbf{v}^T$

The Woodbury formula becomes the Sherman-Morrison identity:

$$(\mathbf{A} + \mathbf{uv}^T)^{-1} = \mathbf{A}^{-1} - \frac{\mathbf{A}^{-1}\mathbf{uv}^T\mathbf{A}^{-1}}{1 + \mathbf{v}^T\mathbf{A}^{-1}\mathbf{u}}$$

Case 3

If we set $\mathbf{A} = \mathbf{I}_p$ and $\mathbf{B} = \mathbf{I}_q$ we have

$$(\mathbf{I}_p + \mathbf{UV})^{-1} = \mathbf{I}_p - \mathbf{U}(\mathbf{I}_q + \mathbf{VU})^{-1} \mathbf{V}$$

Case 4

If we set $\mathbf{A} = \mathbf{I}_p$ $\mathbf{B} = \mathbf{I}_q$ and $q = 1$ we have:

$$(\mathbf{I}_p + \mathbf{uv}^T)^{-1} = \mathbf{I}_p - \frac{\mathbf{uv}^T}{1 + \mathbf{v}^T \mathbf{u}}$$

Studentized residuals (from BECKMAN 1975)

The problem is to find the variance of a generic residual when the Error variance (process variance) is estimated from the experiment.

Let us consider the residual for the i^{th} observation.

The model is $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ with $\boldsymbol{\varepsilon} \sim N_N(\mathbf{0}, \sigma^2 \mathbf{I}_N)$,

the fits are $\hat{\mathbf{Y}} = \mathbf{X}\hat{\boldsymbol{\beta}}$ and the residual vector is $\mathbf{e} = (\mathbf{I} - \mathbf{H})\boldsymbol{\varepsilon}$

As regards the i^{th} residual we have $e_i = y_i - \mathbf{x}_i^T \hat{\boldsymbol{\beta}}$ and its variance is:

$$V(e_i) = \left(1 - \mathbf{x}_i^T (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{x}_i\right) \sigma^2 = (1 - h_{ii}) \sigma^2$$

where

$$h_{ii} = \mathbf{x}_i^T (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{x}_i \quad (1)$$

Is the diagonal element of the Hat Matrix

Considering that the residuals are Normally distributed since they are linear combination of Normal variables we can say that for the generic residual i the standard residual is

$$\frac{e_i}{\sqrt{V(e_i)}} \sim N(0,1) \rightarrow \frac{e_i}{\sqrt{\sigma^2(1-h_{ii})}} \sim N(0,1)$$

If we do not know the error variance σ^2 , we may estimate it with a certain number of degrees of freedom.

Montgomery suggests to use the following relationship for the internal studentized residual:

$$r_i = \frac{e_i}{\sqrt{\frac{\mathbf{e}^T \mathbf{e}}{N-p}(1-h_{ii})}} = \frac{e_i}{\sqrt{\frac{S^2}{N-p}(1-h_{ii})}} \sim t(N-p)$$

Where:

1. $df_E = N - p$ is the number of Error degrees of freedom
2. h_{ii} is the i^{th} diagonal term of the Hat matrix
3. $\mathbf{e}^T \mathbf{e} = S^2$

A different solution is developed in the paper BECKMAN 1974.

Theorem

Let us consider $\mathbf{x}_i^T$, the i^{th} row of the model matrix $\mathbf{X}$ of full rank p .

Rearrange the rows on the matrix $\mathbf{X}$ so that $\mathbf{X}^T = \begin{bmatrix} \mathbf{X}_1^T & \mathbf{x}_i^T \end{bmatrix}$

Let us suppose that the submatrix $\mathbf{X}_1$ is of full rank p .

Let us say $c = \mathbf{x}_i^T (\mathbf{X}_1^T \mathbf{X}_1)^{-1} \mathbf{x}_i$

Then

$$\tau_i = \frac{e_i}{\sqrt{\frac{\mathbf{e}^T \mathbf{e}}{(N-p)}(1+c)}} \rightarrow T = \tau_i \sqrt{\frac{N-p-1}{N-p-\tau_i^2}} \sim t(N-p-1)$$

Proof

$$c = \mathbf{x}_i^T (\mathbf{X}_1^T \mathbf{X}_1)^{-1} \mathbf{x}_i \quad (2)$$

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X} \mathbf{y} \quad (3)$$

$$\hat{\boldsymbol{\beta}}_1 = (\mathbf{X}_1^T \mathbf{X}_1)^{-1} \mathbf{X}_1 \mathbf{y}_1 \quad (4)$$

$$S^2 = \mathbf{e}^T \mathbf{e} = \mathbf{y}^T (\mathbf{I} - \mathbf{H}) \mathbf{y} \quad (5.1)$$

$$S_1^2 = \mathbf{e}_1^T \mathbf{e}_1 = \mathbf{y}_1^T (\mathbf{I} - \mathbf{H}_1) \mathbf{y}_1 \quad (5.2)$$

$$\begin{aligned} \mathbf{X} &= \begin{bmatrix} \mathbf{X}_1 \\ \mathbf{x}_i^T \end{bmatrix} \quad \mathbf{X}^T = \begin{bmatrix} \mathbf{X}_1^T & \mathbf{x}_i \end{bmatrix} \\ \mathbf{X}^T \mathbf{X} &= \mathbf{X}_1^T \mathbf{X}_1 + \mathbf{x}_i \mathbf{x}_i^T \end{aligned} \quad (6)$$

Using the Woodbury formula (case 2)

$$(\mathbf{A} + \mathbf{u} \mathbf{v}^T)^{-1} = \mathbf{A}^{-1} - \frac{\mathbf{A}^{-1} \mathbf{u} \mathbf{v}^T \mathbf{A}^{-1}}{1 + \mathbf{v}^T \mathbf{A}^{-1} \mathbf{u}} \text{ and } \mathbf{A} = \mathbf{X}_1^T \mathbf{X}_1 \text{ and } \mathbf{u} = \mathbf{v} = \mathbf{x}_i \text{ we have}$$

$$\begin{aligned} (\mathbf{X}^T \mathbf{X})^{-1} &= (\mathbf{X}_1^T \mathbf{X}_1)^{-1} - \frac{(\mathbf{X}_1^T \mathbf{X}_1)^{-1} \mathbf{x}_i \mathbf{x}_i^T (\mathbf{X}_1^T \mathbf{X}_1)^{-1}}{1 + \mathbf{x}_i^T (\mathbf{X}_1^T \mathbf{X}_1)^{-1} \mathbf{x}_i} = \\ &= (\mathbf{X}_1^T \mathbf{X}_1)^{-1} - (1+c)^{-1} (\mathbf{X}_1^T \mathbf{X}_1)^{-1} \mathbf{x}_i \mathbf{x}_i^T (\mathbf{X}_1^T \mathbf{X}_1)^{-1} \end{aligned} \quad (7)$$

$$\begin{aligned} \hat{\boldsymbol{\beta}} &= (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y} = \left((\mathbf{X}_1^T \mathbf{X}_1)^{-1} - (1+c)^{-1} (\mathbf{X}_1^T \mathbf{X}_1)^{-1} \mathbf{x}_i \mathbf{x}_i^T (\mathbf{X}_1^T \mathbf{X}_1)^{-1} \right) \begin{bmatrix} \mathbf{X}_1^T & \mathbf{x}_i \end{bmatrix} \begin{bmatrix} \mathbf{y}_1 \\ y_i \end{bmatrix} = \\ &= \left((\mathbf{X}_1^T \mathbf{X}_1)^{-1} - (1+c)^{-1} (\mathbf{X}_1^T \mathbf{X}_1)^{-1} \mathbf{x}_i \mathbf{x}_i^T (\mathbf{X}_1^T \mathbf{X}_1)^{-1} \right) (\mathbf{X}_1^T \mathbf{y}_1 + \mathbf{x}_i y_i) = \\ &= (\mathbf{X}_1^T \mathbf{X}_1)^{-1} \mathbf{X}_1^T \mathbf{y}_1 + (\mathbf{X}_1^T \mathbf{X}_1)^{-1} \mathbf{x}_i y_i - (1+c)^{-1} (\mathbf{X}_1^T \mathbf{X}_1)^{-1} \mathbf{x}_i \mathbf{x}_i^T (\mathbf{X}_1^T \mathbf{X}_1)^{-1} \mathbf{X}_1^T \mathbf{y}_1 - \\ &\quad - (1+c)^{-1} (\mathbf{X}_1^T \mathbf{X}_1)^{-1} \mathbf{x}_i \mathbf{x}_i^T (\mathbf{X}_1^T \mathbf{X}_1)^{-1} \mathbf{x}_i y_i = \\ &= \hat{\boldsymbol{\beta}}_1 - (1+c)^{-1} (\mathbf{X}_1^T \mathbf{X}_1)^{-1} \mathbf{x}_i \mathbf{x}_i^T \hat{\boldsymbol{\beta}}_1 + (\mathbf{X}_1^T \mathbf{X}_1)^{-1} \mathbf{x}_i y_i - (1+c)^{-1} (\mathbf{X}_1^T \mathbf{X}_1)^{-1} \mathbf{x}_i \mathbf{x}_i^T (\mathbf{X}_1^T \mathbf{X}_1)^{-1} \mathbf{x}_i y_i = \\ &= \left[\mathbf{I} - (1+c)^{-1} (\mathbf{X}_1^T \mathbf{X}_1)^{-1} \mathbf{x}_i \mathbf{x}_i^T \right] \hat{\boldsymbol{\beta}}_1 + (\mathbf{X}_1^T \mathbf{X}_1)^{-1} \mathbf{x}_i y_i - (1+c)^{-1} (\mathbf{X}_1^T \mathbf{X}_1)^{-1} \mathbf{x}_i \mathbf{x}_i^T (\mathbf{X}_1^T \mathbf{X}_1)^{-1} \mathbf{x}_i y_i = \\ &= \left[\mathbf{I} - (1+c)^{-1} (\mathbf{X}_1^T \mathbf{X}_1)^{-1} \mathbf{x}_i \mathbf{x}_i^T \right] \hat{\boldsymbol{\beta}}_1 + (1-c(1+c)^{-1}) (\mathbf{X}_1^T \mathbf{X}_1)^{-1} \mathbf{x}_i y_i = \\ &= \left[\mathbf{I} - (1+c)^{-1} (\mathbf{X}_1^T \mathbf{X}_1)^{-1} \mathbf{x}_i \mathbf{x}_i^T \right] \hat{\boldsymbol{\beta}}_1 + (1+c)^{-1} (\mathbf{X}_1^T \mathbf{X}_1)^{-1} \mathbf{x}_i y_i \end{aligned}$$

Hence

$$\hat{\boldsymbol{\beta}} = \left[\mathbf{I} - (1+c)^{-1} (\mathbf{X}_1^T \mathbf{X}_1)^{-1} \mathbf{x}_i \mathbf{x}_i^T \right] \hat{\boldsymbol{\beta}}_1 + (1+c)^{-1} (\mathbf{X}_1^T \mathbf{X}_1)^{-1} \mathbf{x}_i y_i \quad (8)$$

$$\begin{aligned}
 \mathbf{x}_i^T \hat{\boldsymbol{\beta}} &= \mathbf{x}_i^T \left[\mathbf{I} - (1+c)^{-1} (\mathbf{X}_1^T \mathbf{X}_1)^{-1} \mathbf{x}_i \mathbf{x}_i^T \right] \hat{\boldsymbol{\beta}}_1 + (1+c)^{-1} \mathbf{x}_i^T (\mathbf{X}_1^T \mathbf{X}_1)^{-1} \mathbf{x}_i y_i = \\
 &= \mathbf{x}_i^T \left[\mathbf{I} - (1+c)^{-1} (\mathbf{X}_1^T \mathbf{X}_1)^{-1} \mathbf{x}_i \mathbf{x}_i^T \right] \hat{\boldsymbol{\beta}}_1 + c(1+c)^{-1} y_i = \mathbf{x}_i^T \hat{\boldsymbol{\beta}}_1 - c(1+c)^{-1} \mathbf{x}_i^T \hat{\boldsymbol{\beta}}_1 + c(1+c)^{-1} y_i = \\
 &= \left(1 - \frac{c}{1+c} \right) \mathbf{x}_i^T \hat{\boldsymbol{\beta}}_1 + c(1+c)^{-1} y_i = (1+c)^{-1} \mathbf{x}_i^T \hat{\boldsymbol{\beta}}_1 + c(1+c)^{-1} y_i \\
 \mathbf{x}_i^T \hat{\boldsymbol{\beta}} &= (1+c)^{-1} \mathbf{x}_i^T \hat{\boldsymbol{\beta}}_1 + c(1+c)^{-1} y_i \tag{9}
 \end{aligned}$$

Let us prove formulas for S^2 and S_1^2 .

We saw before that:

- a) $\mathbf{X}^T \mathbf{X} = \mathbf{X}_1^T \mathbf{X}_1 + \mathbf{x}_i \mathbf{x}_i^T$
- b) $(\mathbf{X}^T \mathbf{X})^{-1} = (\mathbf{X}_1^T \mathbf{X}_1)^{-1} - (1+c)^{-1} (\mathbf{X}_1^T \mathbf{X}_1)^{-1} \mathbf{x}_i \mathbf{x}_i^T (\mathbf{X}_1^T \mathbf{X}_1)^{-1}$

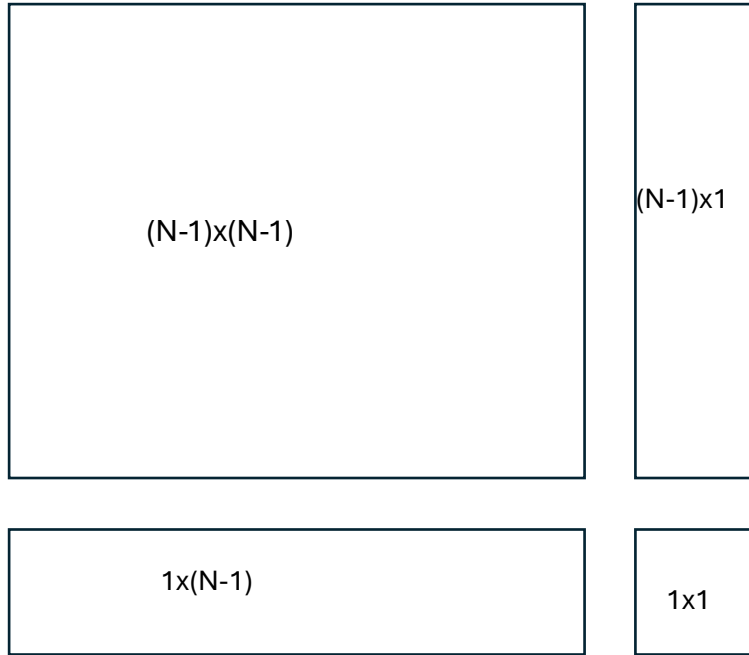
Let $\mathbf{H} = \mathbf{X}(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T$

$$\begin{aligned}
 \mathbf{H} &= \begin{bmatrix} \mathbf{X}_1 \\ \mathbf{x}_i \end{bmatrix} \left[(\mathbf{X}_1^T \mathbf{X}_1)^{-1} - (1+c)^{-1} (\mathbf{X}_1^T \mathbf{X}_1)^{-1} \mathbf{x}_i \mathbf{x}_i^T (\mathbf{X}_1^T \mathbf{X}_1)^{-1} \right] \begin{bmatrix} \mathbf{X}_1^T & \mathbf{x}_i^T \end{bmatrix} = \\
 &= \begin{bmatrix} \mathbf{X}_1 \left[(\mathbf{X}_1^T \mathbf{X}_1)^{-1} - (1+c)^{-1} (\mathbf{X}_1^T \mathbf{X}_1)^{-1} \mathbf{x}_i \mathbf{x}_i^T (\mathbf{X}_1^T \mathbf{X}_1)^{-1} \right] \\ \mathbf{x}_i^T \left[(\mathbf{X}_1^T \mathbf{X}_1)^{-1} - (1+c)^{-1} (\mathbf{X}_1^T \mathbf{X}_1)^{-1} \mathbf{x}_i \mathbf{x}_i^T (\mathbf{X}_1^T \mathbf{X}_1)^{-1} \right] \end{bmatrix} \begin{bmatrix} \mathbf{X}_1^T & \mathbf{x}_i^T \end{bmatrix} = \\
 &= \begin{bmatrix} \mathbf{X}_1 \left[(\mathbf{X}_1^T \mathbf{X}_1)^{-1} - (1+c)^{-1} (\mathbf{X}_1^T \mathbf{X}_1)^{-1} \mathbf{x}_i \mathbf{x}_i^T (\mathbf{X}_1^T \mathbf{X}_1)^{-1} \right] \mathbf{X}_1^T & \mathbf{X}_1 \left[(\mathbf{X}_1^T \mathbf{X}_1)^{-1} - (1+c)^{-1} (\mathbf{X}_1^T \mathbf{X}_1)^{-1} \mathbf{x}_i \mathbf{x}_i^T (\mathbf{X}_1^T \mathbf{X}_1)^{-1} \right] \mathbf{x}_i^T \\ \mathbf{x}_i^T \left[(\mathbf{X}_1^T \mathbf{X}_1)^{-1} - (1+c)^{-1} (\mathbf{X}_1^T \mathbf{X}_1)^{-1} \mathbf{x}_i \mathbf{x}_i^T (\mathbf{X}_1^T \mathbf{X}_1)^{-1} \right] \mathbf{X}_1^T & \mathbf{x}_i^T \left[(\mathbf{X}_1^T \mathbf{X}_1)^{-1} - (1+c)^{-1} (\mathbf{X}_1^T \mathbf{X}_1)^{-1} \mathbf{x}_i \mathbf{x}_i^T (\mathbf{X}_1^T \mathbf{X}_1)^{-1} \right] \mathbf{x}_i^T \end{bmatrix} = \\
 &= \begin{bmatrix} \mathbf{X}_1 (\mathbf{X}_1^T \mathbf{X}_1)^{-1} \mathbf{X}_1^T - (1+c)^{-1} \mathbf{X}_1 (\mathbf{X}_1^T \mathbf{X}_1)^{-1} \mathbf{x}_i \mathbf{x}_i^T (\mathbf{X}_1^T \mathbf{X}_1)^{-1} \mathbf{X}_1^T & \mathbf{X}_1 (\mathbf{X}_1^T \mathbf{X}_1)^{-1} \mathbf{x}_i - c(1+c)^{-1} \mathbf{X}_1 (\mathbf{X}_1^T \mathbf{X}_1)^{-1} \mathbf{x}_i \\ \mathbf{x}_i^T (\mathbf{X}_1^T \mathbf{X}_1)^{-1} \mathbf{X}_1^T - c(1+c)^{-1} \mathbf{x}_i^T (\mathbf{X}_1^T \mathbf{X}_1)^{-1} \mathbf{X}_1^T & c - c^2(1+c)^{-1} \end{bmatrix} = \\
 &= \begin{bmatrix} \mathbf{X}_1 (\mathbf{X}_1^T \mathbf{X}_1)^{-1} \mathbf{X}_1^T - (1+c)^{-1} \mathbf{X}_1 (\mathbf{X}_1^T \mathbf{X}_1)^{-1} \mathbf{x}_i \mathbf{x}_i^T (\mathbf{X}_1^T \mathbf{X}_1)^{-1} \mathbf{X}_1^T & (1-c(1+c)^{-1}) \mathbf{X}_1 (\mathbf{X}_1^T \mathbf{X}_1)^{-1} \mathbf{x}_i \\ (1-c(1+c)^{-1}) \mathbf{x}_i^T (\mathbf{X}_1^T \mathbf{X}_1)^{-1} \mathbf{X}_1^T & c(1+c)^{-1} \end{bmatrix} = \\
 &= \begin{bmatrix} \mathbf{X}_1 (\mathbf{X}_1^T \mathbf{X}_1)^{-1} \mathbf{X}_1^T - (1+c)^{-1} \mathbf{X}_1 (\mathbf{X}_1^T \mathbf{X}_1)^{-1} \mathbf{x}_i \mathbf{x}_i^T (\mathbf{X}_1^T \mathbf{X}_1)^{-1} \mathbf{X}_1^T & (1+c)^{-1} \mathbf{X}_1 (\mathbf{X}_1^T \mathbf{X}_1)^{-1} \mathbf{x}_i \\ (1+c)^{-1} \mathbf{x}_i^T (\mathbf{X}_1^T \mathbf{X}_1)^{-1} \mathbf{X}_1^T & c(1+c)^{-1} \end{bmatrix}
 \end{aligned}$$

Summing up

$$\mathbf{H} = \begin{bmatrix} \mathbf{H}_1 - (1+c)^{-1} \mathbf{X}_1 (\mathbf{X}_1^T \mathbf{X}_1)^{-1} \mathbf{x}_i \mathbf{x}_i^T (\mathbf{X}_1^T \mathbf{X}_1)^{-1} \mathbf{X}_1^T & (1+c)^{-1} \mathbf{X}_1 (\mathbf{X}_1^T \mathbf{X}_1)^{-1} \mathbf{x}_i \\ (1+c)^{-1} \mathbf{x}_i^T (\mathbf{X}_1^T \mathbf{X}_1)^{-1} \mathbf{X}_1^T & c(1+c)^{-1} \end{bmatrix} \tag{10}$$

The dimensions of the $\mathbf{H}$ matrix submatrices are sketched in the following figure



Let us develop S^2 and S_1^2

$$\begin{aligned}
 S^2 &= \mathbf{y}^T (\mathbf{I} - \mathbf{H}) \mathbf{y} = \\
 &= \begin{bmatrix} \mathbf{y}_1^T \\ y_i \end{bmatrix}^T \left(\mathbf{I} - \begin{bmatrix} \mathbf{H}_1 - (1+c)^{-1} \mathbf{X}_1 (\mathbf{X}_1^T \mathbf{X}_1)^{-1} \mathbf{x}_i \mathbf{x}_i^T (\mathbf{X}_1^T \mathbf{X}_1)^{-1} \mathbf{X}_1^T & (1+c)^{-1} \mathbf{X}_1 (\mathbf{X}_1^T \mathbf{X}_1)^{-1} \mathbf{x}_i \\ (1+c)^{-1} \mathbf{x}_i^T (\mathbf{X}_1^T \mathbf{X}_1)^{-1} \mathbf{X}_1^T & c(1+c)^{-1} \end{bmatrix} \right) \begin{bmatrix} \mathbf{y}_1 \\ y_i \end{bmatrix} = \\
 &= \begin{bmatrix} \mathbf{y}_1^T & y_i \end{bmatrix} \left(\begin{bmatrix} \mathbf{I}_{N-1} & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} \mathbf{H}_1 - (1+c)^{-1} \mathbf{X}_1 (\mathbf{X}_1^T \mathbf{X}_1)^{-1} \mathbf{x}_i \mathbf{x}_i^T (\mathbf{X}_1^T \mathbf{X}_1)^{-1} \mathbf{X}_1^T & (1+c)^{-1} \mathbf{X}_1 (\mathbf{X}_1^T \mathbf{X}_1)^{-1} \mathbf{x}_i \\ (1+c)^{-1} \mathbf{x}_i^T (\mathbf{X}_1^T \mathbf{X}_1)^{-1} \mathbf{X}_1^T & c(1+c)^{-1} \end{bmatrix} \right) \begin{bmatrix} \mathbf{y}_1 \\ y_i \end{bmatrix} = \\
 &= \begin{bmatrix} \mathbf{y}_1^T & y_i \end{bmatrix} \begin{bmatrix} \mathbf{I}_{N-1} - \left(\mathbf{H}_1 - (1+c)^{-1} \mathbf{X}_1 (\mathbf{X}_1^T \mathbf{X}_1)^{-1} \mathbf{x}_i \mathbf{x}_i^T (\mathbf{X}_1^T \mathbf{X}_1)^{-1} \mathbf{X}_1^T \right) & -(1+c)^{-1} \mathbf{X}_1 (\mathbf{X}_1^T \mathbf{X}_1)^{-1} \mathbf{x}_i \\ -(1+c)^{-1} \mathbf{x}_i^T (\mathbf{X}_1^T \mathbf{X}_1)^{-1} \mathbf{X}_1^T & 1 - c(1+c)^{-1} \end{bmatrix} \begin{bmatrix} \mathbf{y}_1 \\ y_i \end{bmatrix} = \\
 &= \begin{bmatrix} \mathbf{y}_1^T & y_i \end{bmatrix} \begin{bmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{B}^T & C \end{bmatrix} \begin{bmatrix} \mathbf{y}_1 \\ y_i \end{bmatrix} = \begin{bmatrix} \mathbf{y}_1^T \mathbf{A} + y_i \mathbf{B}^T & \mathbf{y}_1^T \mathbf{B} + y_i C \end{bmatrix} \begin{bmatrix} \mathbf{y}_1 \\ y_i \end{bmatrix} = \\
 &= \mathbf{y}_1^T \mathbf{A} \mathbf{y}_1 + y_i \mathbf{B}^T \mathbf{y}_1 + \mathbf{y}_1^T \mathbf{B} y_i + y_i^2 (1 - c(1+c)^{-1}) = \mathbf{y}_1^T \mathbf{A} \mathbf{y}_1 + y_i \mathbf{B}^T \mathbf{y}_1 + \mathbf{y}_1^T \mathbf{B} y_i + (1+c)^{-1} y_i^2
 \end{aligned}$$

Let us develop the terms of the sum:

$$\begin{aligned}
 \mathbf{y}_1^T \mathbf{A} \mathbf{y}_1 &= \mathbf{y}_1^T \left(\mathbf{I}_{N-1} - \left(\mathbf{H}_1 - (1+c)^{-1} \mathbf{X}_1 (\mathbf{X}_1^T \mathbf{X}_1)^{-1} \mathbf{x}_i \mathbf{x}_i^T (\mathbf{X}_1^T \mathbf{X}_1)^{-1} \mathbf{X}_1^T \right) \right) \mathbf{y}_1 = \\
 &= \mathbf{y}_1^T (\mathbf{I}_{N-1} - \mathbf{H}_1) \mathbf{y}_1 + (1+c)^{-1} \mathbf{y}_1^T \mathbf{X}_1 (\mathbf{X}_1^T \mathbf{X}_1)^{-1} \mathbf{x}_i \mathbf{x}_i^T (\mathbf{X}_1^T \mathbf{X}_1)^{-1} \mathbf{X}_1^T \mathbf{y}_1 = \\
 &= S_1^2 + (1+c)^{-1} \hat{\boldsymbol{\beta}}_1^T \mathbf{x}_i \mathbf{x}_i^T \hat{\boldsymbol{\beta}}_1
 \end{aligned}$$

$$y_i \mathbf{B}^T \mathbf{y}_1 = -y_i (1+c)^{-1} \mathbf{x}_i^T (\mathbf{X}_1^T \mathbf{X}_1)^{-1} \mathbf{X}_1^T \mathbf{y}_1 = -y_i (1+c)^{-1} \mathbf{x}_i^T \hat{\boldsymbol{\beta}}_1$$

$$\mathbf{y}_1^T \mathbf{B} \mathbf{y}_i = -y_i \mathbf{y}_1^T (1+c)^{-1} \mathbf{X}_1 (\mathbf{X}_1^T \mathbf{X}_1)^{-1} \mathbf{x}_i = -y_i (1+c)^{-1} \hat{\boldsymbol{\beta}}_1^T \mathbf{x}_i$$

$$C y_i^2 = (1-c(1+c)^{-1}) y_i^2 = (1+c)^{-1} y_i^2$$

$$\begin{aligned}
 S^2 &= S_1^2 + (1+c)^{-1} \hat{\boldsymbol{\beta}}_1^T \mathbf{x}_i \mathbf{x}_i^T \hat{\boldsymbol{\beta}}_1 - y_i (1+c)^{-1} \mathbf{x}_i^T \hat{\boldsymbol{\beta}}_1 - y_i (1+c)^{-1} \hat{\boldsymbol{\beta}}_1^T \mathbf{x}_i + (1+c)^{-1} y_i^2 = \\
 &= S_1^2 + (1+c)^{-1} \left(\hat{\boldsymbol{\beta}}_1^T \mathbf{x}_i \mathbf{x}_i^T \hat{\boldsymbol{\beta}}_1 - y_i \mathbf{x}_i^T \hat{\boldsymbol{\beta}}_1 - y_i \hat{\boldsymbol{\beta}}_1^T \mathbf{x}_i + y_i^2 \right) = \\
 &= S_1^2 + (1+c)^{-1} \left(y_i (y_i - \hat{\boldsymbol{\beta}}_1^T \mathbf{x}_i) - \mathbf{x}_i^T \hat{\boldsymbol{\beta}}_1 (y_i - \hat{\boldsymbol{\beta}}_1^T \mathbf{x}_i) \right) = \\
 &= S_1^2 + (1+c)^{-1} \left(y_i - \mathbf{x}_i^T \hat{\boldsymbol{\beta}}_1 \right)^2
 \end{aligned}$$

$$\text{since } \mathbf{x}_i^T \hat{\boldsymbol{\beta}}_1 = \hat{\boldsymbol{\beta}}_1^T \mathbf{x}_i$$

If we consider that

$$\mathbf{x}_i^T \hat{\boldsymbol{\beta}} = (1+c)^{-1} \mathbf{x}_i^T \hat{\boldsymbol{\beta}}_1 + c(1+c)^{-1} y_i \rightarrow \mathbf{x}_i^T \hat{\boldsymbol{\beta}} = (1+c) \mathbf{x}_i^T \hat{\boldsymbol{\beta}} - c y_i$$

We can write:

$$\begin{aligned}
 S^2 &= S_1^2 + (1+c)^{-1} \left(y_i - \mathbf{x}_i^T \hat{\boldsymbol{\beta}} \right)^2 = S_1^2 + (1+c)^{-1} \left(y_i - (1+c) \mathbf{x}_i^T \hat{\boldsymbol{\beta}} + c y_i \right)^2 = \\
 &= S_1^2 + (1+c)^{-1} (1+c)^2 \left(y_i - \mathbf{x}_i^T \hat{\boldsymbol{\beta}} \right)^2 = S_1^2 + (1+c) \left(y_i - \mathbf{x}_i^T \hat{\boldsymbol{\beta}} \right)^2
 \end{aligned}$$

If we say

$$e_{(i)} = \left(y_i - \mathbf{x}_i^T \hat{\boldsymbol{\beta}} \right) \quad \text{and} \quad e_i = \left(y_i - \mathbf{x}_i^T \hat{\boldsymbol{\beta}} \right)$$

We have:

$$S^2 = S_1^2 + (1+c)^{-1} e_{(i)}^2 \quad (11)$$

or

$$S^2 = S_1^2 + (1+c) e_i^2 \quad (12)$$

From these equalities we can write

$$\begin{aligned}
 (1+c) e_i^2 &= (1+c)^{-1} e_{(i)}^2 \\
 e_{(i)} &= (1+c) e_i
 \end{aligned} \quad (13)$$

From $(\mathbf{X}^T \mathbf{X})^{-1} = (\mathbf{X}_1^T \mathbf{X}_1)^{-1} - (1+c)^{-1} (\mathbf{X}_1^T \mathbf{X}_1)^{-1} \mathbf{x}_i \mathbf{x}_i^T (\mathbf{X}_1^T \mathbf{X}_1)^{-1}$ we have

$$\mathbf{x}_i^T (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{x}_i = \mathbf{x}_i^T (\mathbf{X}_1^T \mathbf{X}_1)^{-1} \mathbf{x}_i - (1+c)^{-1} \mathbf{x}_i^T (\mathbf{X}_1^T \mathbf{X}_1)^{-1} \mathbf{x}_i \mathbf{x}_i^T (\mathbf{X}_1^T \mathbf{X}_1)^{-1} \mathbf{x}_i = c - (1+c)^{-1} c^2 = c(1+c)^{-1}$$

or

$$h_{ii} = c(1+c)^{-1} \quad (14)$$

$$1 - h_{ii} = (1+c)^{-1} \quad (15)$$

And

$$e_{(i)} = \frac{e_i}{1 - h_{ii}} \quad (16)$$

If the model is unbiased and considering that y_i and $\mathbf{x}_i^T \hat{\boldsymbol{\beta}}_1$ are independent, since $\hat{\boldsymbol{\beta}}_1$ is estimated without y_i , we have:

$$E(y_i - \mathbf{x}_i^T \hat{\boldsymbol{\beta}}_1) = 0$$

$$V(y_i - \mathbf{x}_i^T \hat{\boldsymbol{\beta}}_1) = V(y_i) + V(\mathbf{x}_i^T \hat{\boldsymbol{\beta}}_1) = \sigma^2 + \mathbf{x}_i^T (\mathbf{X}_1^T \mathbf{X}_1)^{-1} \mathbf{x}_i \sigma^2 = (1+c) \sigma^2$$

Hence if we substitute the variance with its estimate, we have the external studentized residual:

$$\tau_i = \frac{y_i - \mathbf{x}_i^T \hat{\boldsymbol{\beta}}_1}{\sqrt{\frac{S_1^2}{N-p-1} (1+c)}} \sim t(N-p-1)$$

Using (9) and (12) we have

$$\tau_i = \frac{y_i - \mathbf{x}_i^T \hat{\boldsymbol{\beta}}_1}{\sqrt{\frac{S_1^2}{N-p-1} (1+c)}} = \frac{(\mathbf{y} - \mathbf{X}_1^T \hat{\boldsymbol{\beta}}_1)(1+c)}{\sqrt{\frac{S^2 - (1+c)e_i^2}{N-p-1} (1+c)}} = \frac{e_i}{\sqrt{\frac{S^2 - (1+c)e_i^2}{(N-p-1)(1+c)}}}$$

The studentized residual r_i can be written

$$r_i = \frac{e_i}{\sqrt{\frac{S^2}{N-p} (1-h_{ii})}} = \frac{e_i}{\sqrt{\frac{S^2}{N-p} (1+c)^{-1}}} \text{ and } T = \frac{r_i}{\sqrt{\frac{N-p-r_i^2}{N-p-1}}} \sim t(N-p-1)$$

External Studentized Residuals

To studentized the residual i , we must **not** use the observation i to compute the estimated variance. The formula is:

$$\tau_i = \frac{e_i}{\sqrt{s_{(i)}^2 (1-h_{ii})}} \sim t(N-p-1)$$

Where

$$s_{(i)}^2 = \frac{SS_E - SS(e_i)}{(N-p)-1} = \frac{(N-p)MS_E - e_i^2 / (1-h_{ii})}{N-p-1}$$

Proof that $SS(e_i) = \frac{e_i^2}{1-h_{ii}}$

We start by finding the appropriate Sum of Squares of the set of linear equations $\mathbf{C}^T \mathbf{y}$ where $\mathbf{C}$ is a matrix $N \times m$, $\mathbf{y}$ is a vector $N \times 1$.

The appropriate Sum of Squares is the square on the length of the orthogonal projection of the vector $\mathbf{y}$ onto the linear space M spanned by the columns of the matrix $\mathbf{C}$.

We use the same approach used in regression:

Let us define $\mathbf{Cz}$ the general vector belonging to the linear space M .

The orthogonal projection is found by the solution of the following equation $\mathbf{C}^T \mathbf{C} \hat{\mathbf{z}} = \mathbf{C}^T \mathbf{y}$

We have that $\hat{\mathbf{z}} = (\mathbf{C}^T \mathbf{C})^{-1} \mathbf{C}^T \mathbf{y}$ if the inverse exists, if the inverse does not exist, the projection onto the space will exist (imagine it by intuitive geometry) and it will be unique (It is found using the concept of Generalized Inverse).

Now we can find the associated Sum of Squares (the same formula we used in regression):

$$SS(\mathbf{C}^T \mathbf{y}) = SS(\mathbf{C} \hat{\mathbf{z}}) = \hat{\mathbf{z}}^T \mathbf{C}^T \mathbf{y} = \mathbf{y}^T \mathbf{C} \hat{\mathbf{z}}$$

Special case no.1

Let us suppose that $\mathbf{C} \rightarrow \mathbf{c}$ where $\dim(\mathbf{c}) = N \cdot 1$ we have that: $\hat{z} = (\mathbf{c}^T \mathbf{c})^{-1} \mathbf{c}^T \mathbf{y} = \frac{\mathbf{c}^T \mathbf{y}}{\mathbf{c}^T \mathbf{c}}$ since the

inverse of a constant exists and $SS(\mathbf{c}^T \mathbf{y}) = SS(\mathbf{c} \hat{z}) = \hat{z}^T \mathbf{c}^T \mathbf{y} = \frac{(\mathbf{c}^T \mathbf{y})^2}{\mathbf{c}^T \mathbf{c}}$

Now let us consider the i^{th} residual:

$$e_i = -h_{i1}y_1 - h_{i2}y_2, \dots, + (1 - h_{ii})y_i, \dots - h_{iN}y_N = \mathbf{c}_i^T \mathbf{y}$$

$$\mathbf{c}_i^T \mathbf{c}_i = \sum_{\substack{l=1, N \\ l \neq i}} h_{il}^2 + (1 - h_{ii})^2 = \sum_{\substack{l=1, N \\ l \neq i}} h_{il}^2 + 1 - 2h_{ii} + h_{ii}^2 = \sum_{l=1, N} h_{il}^2 + 1 - 2h_{ii}$$

and considering the idempotence property of $\mathbf{H}$, $\sum_{l=1, N} h_{il}^2 = h_{ii}$, we have $\mathbf{c}_i^T \mathbf{c}_i = \sum_{l=1, N} h_{il}^2 + 1 - 2h_{ii} = 1 - h_{ii}$

$$\text{and, eventually, } SS(e_i) = \frac{(\mathbf{c}_i^T \mathbf{y})^2}{\mathbf{c}_i^T \mathbf{c}_i} = \frac{e_i^2}{(1 - h_{ii})}$$

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Expected value of MS_{LOF} (Lack of Fit)

Nomenclature and assumptions

$$\boldsymbol{\varepsilon} \sim N(\mathbf{b}, \mathbf{I}_N \sigma^2) \quad \mathbf{b} = \text{Bias}$$

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon} \rightarrow E(\mathbf{y}) = \mathbf{X}\boldsymbol{\beta} + \mathbf{b} \quad \text{or} \quad \mathbf{b} = E(\mathbf{y}) - \mathbf{X}\boldsymbol{\beta}$$

$$E(MS_{LOF}) = \frac{E(SS_{LOF})}{m - p}$$

$$SS_E = SS_{PE} + SS_{LOF} \rightarrow SS_{LOF} = SS_E - SS_{PE}$$

$$E(SS_{LOF}) = E(SS_E) - E(SS_{PE})$$

$$E(SS_{PE}) = \sum_{i=1, m} (n_i - 1) E(s_i^2) = (N - m) \sigma^2$$

$$\begin{aligned}
 E(SS_E) &= E(\mathbf{e}^T \mathbf{e}) = E(\mathbf{y}^T (\mathbf{I} - \mathbf{H}) \mathbf{y}) = E(\mathbf{y})^T (\mathbf{I} - \mathbf{H}) E(\mathbf{y}) + \text{Tr}[(\mathbf{I} - \mathbf{H}) \Sigma_y] = \\
 &= (\mathbf{X}\boldsymbol{\beta} + \mathbf{b})^T (\mathbf{I} - \mathbf{H}) (\mathbf{X}\boldsymbol{\beta} + \mathbf{b}) + \text{Tr}[(\mathbf{I} - \mathbf{H}) \mathbf{I} \sigma^2] = (\boldsymbol{\beta}^T \mathbf{X}^T + \mathbf{b}^T) (\mathbf{I} - \mathbf{H}) (\mathbf{X}\boldsymbol{\beta} + \mathbf{b}) + \sigma^2 \text{Tr}(\mathbf{I} - \mathbf{H}) = \\
 &= \boldsymbol{\beta}^T \mathbf{X}^T (\mathbf{I} - \mathbf{H}) \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\beta}^T \mathbf{X}^T (\mathbf{I} - \mathbf{H}) \mathbf{b} + \mathbf{b}^T (\mathbf{I} - \mathbf{H}) \mathbf{X}\boldsymbol{\beta} + \mathbf{b}^T (\mathbf{I} - \mathbf{H}) \mathbf{b} + \sigma^2 (N - p) = \\
 &= \boldsymbol{\beta}^T \mathbf{X}^T (\mathbf{I} - \mathbf{H}) \mathbf{X}\boldsymbol{\beta} + 2\mathbf{b}^T (\mathbf{I} - \mathbf{H}) \mathbf{X}\boldsymbol{\beta} + \mathbf{b}^T (\mathbf{I} - \mathbf{H}) \mathbf{b} + \sigma^2 (N - p)
 \end{aligned}$$

$$a) \boldsymbol{\beta}^T \mathbf{X}^T (\mathbf{I} - \mathbf{H}) \mathbf{X}\boldsymbol{\beta} = \boldsymbol{\beta}^T \mathbf{X}^T \mathbf{X}\boldsymbol{\beta} - \boldsymbol{\beta}^T \mathbf{X}^T \mathbf{H} \mathbf{X}\boldsymbol{\beta} = \boldsymbol{\beta}^T \mathbf{X}^T \mathbf{X}\boldsymbol{\beta} - \boldsymbol{\beta}^T \mathbf{X}^T \mathbf{X} (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{X}\boldsymbol{\beta} = 0$$

$$b) 2\mathbf{b}^T (\mathbf{I} - \mathbf{H}) \mathbf{X}\boldsymbol{\beta} = 2\mathbf{b}^T \mathbf{X}\boldsymbol{\beta} - 2\mathbf{b}^T \mathbf{H} \mathbf{X}\boldsymbol{\beta} = 2\mathbf{b}^T \mathbf{X}\boldsymbol{\beta} - 2\mathbf{b}^T \mathbf{X} (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{X}\boldsymbol{\beta} = 0$$

And

$$E(SS_E) = \mathbf{b}^T (\mathbf{I} - \mathbf{H}) \mathbf{b} + \sigma^2 (N - p) = (E(\mathbf{y}) - \mathbf{X}\boldsymbol{\beta})^T (\mathbf{I} - \mathbf{H}) (E(\mathbf{y}) - \mathbf{X}\boldsymbol{\beta}) + \sigma^2 (N - p)$$

$$E(MS_{LOF}) = \frac{E(SS_{LOF})}{m - p} = \frac{E(SS_E) - E(SS_{PE})}{m - p} = \frac{(E(\mathbf{y}) - \mathbf{X}\boldsymbol{\beta})^T (\mathbf{I} - \mathbf{H}) (E(\mathbf{y}) - \mathbf{X}\boldsymbol{\beta}) + \sigma^2 (N - p) - (N - m) \sigma^2}{m - p}$$

In conclusion we have

$$E(MS_{LOF}) = \sigma^2 + \frac{(E(\mathbf{y}) - \mathbf{X}\boldsymbol{\beta})^T (\mathbf{I} - \mathbf{H}) (E(\mathbf{y}) - \mathbf{X}\boldsymbol{\beta})}{m - p}$$

Extra Sum of Squares Principle: General proof

Let us consider the Full Model and the Reduced Model:

Full Model $\mathbf{y} = \mathbf{X}_1\boldsymbol{\beta}_1 + \mathbf{X}_2\boldsymbol{\beta}_2 + \boldsymbol{\varepsilon}$

Reduced Model $\mathbf{y} = \mathbf{X}_2\boldsymbol{\beta}_2 + \boldsymbol{\varepsilon}$

Let us project the matrix $\mathbf{X}_1$ on $\mathbf{X}_2$ and compute the residuals

$$\mathbf{X}_{1,2} = \mathbf{X}_1 - \mathbf{X}_2(\mathbf{X}_2^T\mathbf{X}_2)^{-1}\mathbf{X}_2^T\mathbf{X}_1 \rightarrow \mathbf{X}_1 = \mathbf{X}_{1,2} + \mathbf{X}_2(\mathbf{X}_2^T\mathbf{X}_2)^{-1}\mathbf{X}_2^T\mathbf{X}_1$$

The Full Model becomes $\mathbf{y} = (\mathbf{X}_{1,2} + \mathbf{X}_2(\mathbf{X}_2^T\mathbf{X}_2)^{-1}\mathbf{X}_2^T\mathbf{X}_1)\boldsymbol{\beta}_1 + \mathbf{X}_2\boldsymbol{\beta}_2 + \boldsymbol{\varepsilon}$

$$\mathbf{X}_{1,2}\boldsymbol{\beta}_1 + \mathbf{X}_2(\mathbf{X}_2^T\mathbf{X}_2)^{-1}\mathbf{X}_2^T\mathbf{X}_1\boldsymbol{\beta}_1 + \mathbf{X}_2\boldsymbol{\beta}_2 = \mathbf{X}_{1,2}\boldsymbol{\beta}_1 + \mathbf{X}_2(\mathbf{B}\boldsymbol{\beta}_1 + \boldsymbol{\beta}_2)$$

With $\mathbf{B} = (\mathbf{X}_2^T\mathbf{X}_2)^{-1}\mathbf{X}_2^T\mathbf{X}_1$ and $\boldsymbol{\alpha} = \mathbf{B}\boldsymbol{\beta}_1 + \boldsymbol{\beta}_2$ we have $\mathbf{y} = \mathbf{X}_{1,2}\boldsymbol{\beta}_1 + \mathbf{X}_2\boldsymbol{\alpha} + \boldsymbol{\varepsilon}$

Note 1 if $H_0 : \boldsymbol{\beta}_1 = \mathbf{0} \rightarrow \boldsymbol{\alpha} = \boldsymbol{\beta}_2$

Note 2: The two parts of the model are orthogonal:

$$\mathbf{X}_{1,2}^T\mathbf{X}_2 = (\mathbf{X}_1 - \mathbf{X}_2(\mathbf{X}_2^T\mathbf{X}_2)^{-1}\mathbf{X}_2^T\mathbf{X}_1)^T\mathbf{X}_2 = \mathbf{X}_1^T\mathbf{X}_2 - \mathbf{X}_1^T\mathbf{X}_2(\mathbf{X}_2^T\mathbf{X}_2)^{-1}\mathbf{X}_2^T\mathbf{X}_2 = \mathbf{0} = \mathbf{X}_2^T\mathbf{X}_{1,2}$$

The test hypotheses are : $H_0 : \boldsymbol{\beta}_1 = \mathbf{0}$ vs $H_1 : \boldsymbol{\beta}_1 \neq \mathbf{0}$

The estimate of the model parameters is :

$$\begin{aligned} \begin{Bmatrix} \hat{\boldsymbol{\beta}}_1 \\ \hat{\boldsymbol{\alpha}} \end{Bmatrix} &= \left([\mathbf{X}_{1,2} \ \mathbf{X}_2]^T [\mathbf{X}_{1,2} \ \mathbf{X}_2] \right)^{-1} [\mathbf{X}_{1,2} \ \mathbf{X}_2]^T \mathbf{y} = \begin{bmatrix} \mathbf{X}_{1,2}^T\mathbf{X}_{1,2} & 0 \\ 0 & \mathbf{X}_2^T\mathbf{X}_2 \end{bmatrix}^{-1} \begin{bmatrix} \mathbf{X}_{1,2}^T\mathbf{y} \\ \mathbf{X}_2^T\mathbf{y} \end{bmatrix} = \\ &= \begin{Bmatrix} (\mathbf{X}_{1,2}^T\mathbf{X}_{1,2})^{-1}\mathbf{X}_{1,2}^T\mathbf{y} \\ (\mathbf{X}_2^T\mathbf{X}_2)^{-1}\mathbf{X}_2^T\mathbf{y} \end{Bmatrix} \rightarrow \begin{cases} \hat{\boldsymbol{\beta}}_1 = (\mathbf{X}_{1,2}^T\mathbf{X}_{1,2})^{-1}\mathbf{X}_{1,2}^T\mathbf{y} \\ \hat{\boldsymbol{\alpha}} = (\mathbf{X}_2^T\mathbf{X}_2)^{-1}\mathbf{X}_2^T\mathbf{y} \end{cases} \end{aligned}$$

The partition of the Sum of Squares

Full Model: $SS_{REG}^*(F) = \begin{Bmatrix} \hat{\boldsymbol{\beta}}_1 \\ \hat{\boldsymbol{\alpha}} \end{Bmatrix}^T [\mathbf{X}_{1,2} \ \mathbf{X}_2]^T \mathbf{y} = \hat{\boldsymbol{\beta}}_1^T \mathbf{X}_{1,2}^T \mathbf{y} + \hat{\boldsymbol{\alpha}}^T \mathbf{X}_2^T \mathbf{y} \quad df_{REG}^*(F) = p$

Reduced Model: $SS_{REG}^*(R) = \hat{\boldsymbol{\alpha}}^T \mathbf{X}_2^T \mathbf{y} \quad df_{REG}^*(R) = p - r$

Or we can write $SS_{REG}^*(F) = SS_{REG}^*(R) + \hat{\boldsymbol{\beta}}_1^T \mathbf{X}_{1,2}^T \mathbf{y}$ or

$$SS_{REG}^*(F) - SS_{REG}^*(R) = \hat{\boldsymbol{\beta}}_1^T \mathbf{X}_{1,2}^T \mathbf{y} = \hat{\boldsymbol{\beta}}_1^T \mathbf{X}_{1,2}^T \mathbf{X}_{1,2} \hat{\boldsymbol{\beta}}_1 = \hat{\boldsymbol{\beta}}_1^T \Sigma_{\hat{\boldsymbol{\beta}}_1}^{-1} \hat{\boldsymbol{\beta}}_1 \sigma^2$$

With $\hat{\boldsymbol{\beta}}_1 = (\mathbf{X}_{1,2}^T\mathbf{X}_{1,2})^{-1}\mathbf{X}_{1,2}^T\mathbf{y}$ and $V(\hat{\boldsymbol{\beta}}_1) \equiv \Sigma_{\hat{\boldsymbol{\beta}}_1} = (\mathbf{X}_{1,2}^T\mathbf{X}_{1,2})^{-1} \sigma^2$

If H_0 holds $E(\hat{\boldsymbol{\beta}}_1) = \boldsymbol{\beta}_1 = \mathbf{0}$ and $\frac{SS_{REG}^*(F) - SS_{REG}^*(R)}{\sigma^2} = \hat{\boldsymbol{\beta}}_1^T \Sigma_{\hat{\boldsymbol{\beta}}_1}^{-1} \hat{\boldsymbol{\beta}}_1 \sim \chi^2(r)$

And: $\frac{SS_{REG}^*(F) - SS_{REG}^*(R)}{r\sigma^2} = \frac{SS_{REG}^*(\hat{\boldsymbol{\beta}}_1 | \hat{\boldsymbol{\beta}}_2)}{r\sigma^2} \sim \frac{\chi^2(r)}{r}$

We know that: $\frac{MS_E}{\sigma^2} \sim \frac{\chi^2(N-p)}{N-p}$

in conclusion $F_0 = \frac{SS_{REG}^*(\hat{\boldsymbol{\beta}}_1 | \hat{\boldsymbol{\beta}}_2)/r}{MS_E} = \frac{\chi^2(r)/r}{\chi^2(N-p)/(N-p)} \sim F(r, N-p)$

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References

[1] Beckman R. J. Trussel H. J. "The distribution of an Arbitrary Studentized Residual and the Effects of Updating in Multiple Regression" Journal of the American Statistical Association, March 1974, Volume 63, Number 345.